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17/100

1. a) data mining:

data mining is the process of extracting useful pattern and knowledge from large data base.

two data-mining tasks: i) classification
ii) clustring

KDD Process:
i) data cleaning
ii) data integration
iii) data selection
iv) data transformation
v) data mining
vi) pattern evaluation
vii) knowledge presentation.

⇒ Remove errors and missing values.

⇒ improve data quality and mining accuracy.

2. a. data integration; data integration combines data from multiple sources into one data set.

two challenges: * data redunancy

* data inconsistency

* handling redundancy: Removes duplicate records.

* handling inconsistency;

standaridize data formats and values

b) covaraince analysis:

covariance measures how two variables change together.

i. population covarience $cov(x,y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n}$

3. a) stocks A and B

covariance = +5 [positive]

=> the stock prices rise together

=> they are positively corelated and not indendent.

b) data mining system architecture.

components : * data sources

* database / datawarehouse

* database server

* data mining engine

* pattern evaluation

* knowledge base.

Role : these components work together to discover useful knowledge from date.

7. a) given

$x = \{4, 5, 6, 7, 9\}$

$y = \{8, 3, 4, 5, 6\}$

co variance = -0.45

interpretation : Negative Co-varience means that vaiable movy in opposite direction.

b) data preprocessing:

Major steps

i. data cleaning

ii. data integration

iii data transformation

iv. data reduction.

Pipeline:

Raw data → cleaning → integration → transforma-tion → prepared data → discreatization.

5\. a. equal. Frequency bining: equal frequency bining divides data into bin with an equal number of values.

For 27, values, using 3 bins:

* Bin 1: 13-22 [9 valus]
* Bin 2: 22-35 [9 valus]
* Bin 3: 35-70 [9 valus]

b. data discreatization

data discretization converts continuos data into intervals [bins].

Importance:

* Reduces noise
* Simplicities analysis
* improves mining performance.